

Data Profile & Processing

Complaint → Regulation Classifier (CMPL-REG-24)
Curated CFPB consumer-complaint narratives

Generated: 2026-07-21 15:07 UTC
Source: CFPB Consumer Complaint Database (public)
Dataset: data/complaints/cfpb_complaints.csv - 4,000 rows
Model: CMPL-REG-24 - repo: reg-agents

1 · Processing pipeline

stage	what it does	parameter / evidence
1 · Acquire	Download from the public CFPB Cor	scripts/fetch_cfpb_complaints.py parse of the bu
2 · Length filter	Keep narratives $\geq$ 120 chars; truncat	MIN_CHARS=120 · MAX_CHARS=1800
3 · Exact dedup	Hash of whitespace/case-normalize	0 duplicates remain (verified below)
4 · Near dedup	First-200-chars normalized fingerpr	removes boilerplate template complaints
5 · PII check	CFPB pre-masks PII as 'XXXX'; verifi	82% of narratives carry masking tokens
6 · Balanced sampling	Per product-issue cap to fight credit	PER_ISSUE_CAP=350 · max observed 126
7 · Weak labeling	CFPB product/issue taxonomy + key	reg_agents/common/complaints.py REGULATORY
8 · Split	75/25 train/test, stratified on is_reg	train 3,000 / test 1,000

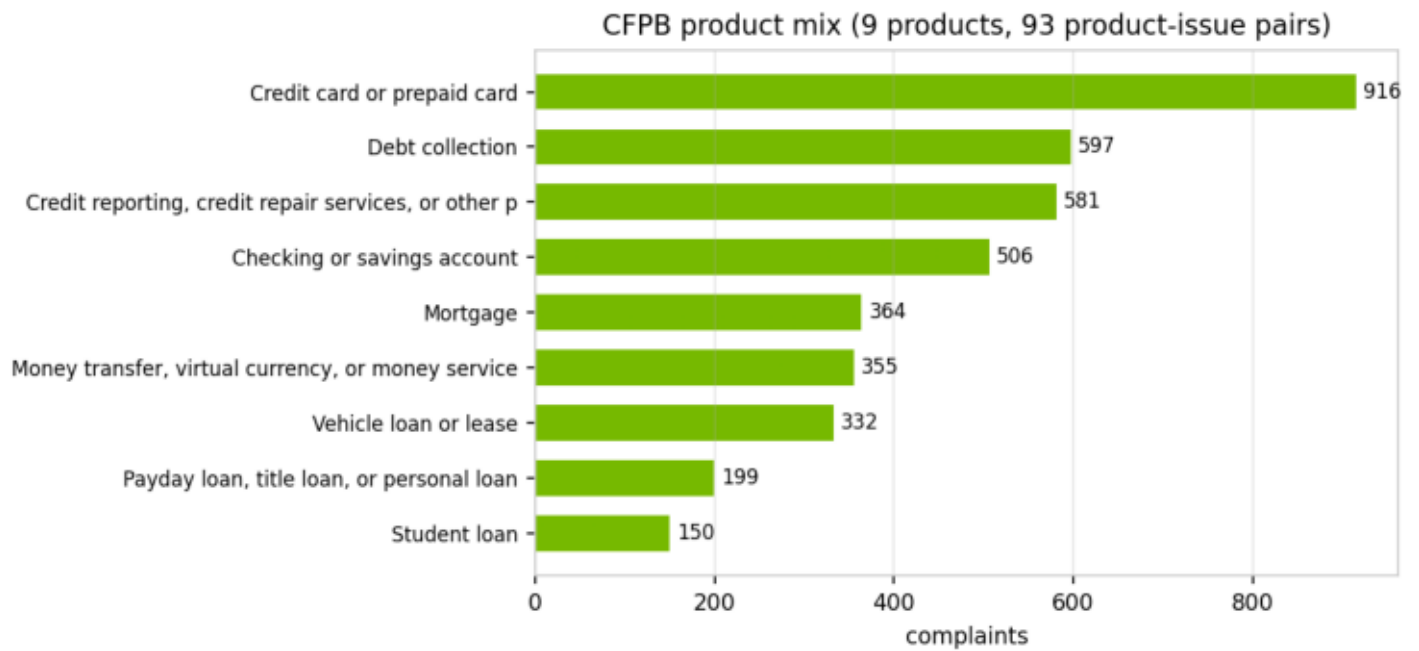
2 · Dataset schema

column	dtype	non-null	unique
complaint_id	int64	4,000	4,000
date_received	object	4,000	67
product	object	4,000	9
sub_product	object	4,000	47
issue	object	4,000	73
sub_issue	object	2,970	146
company	object	4,000	557
state	object	3,986	56
tags	object	769	3
narrative	object	4,000	4,000
label	object	4,000	24
is_regulatory	int64	4,000	2

3 · Composition

The dataset is NOT all-regulatory: 3,865 of 4,000 narratives (96.6%) carry a weak regulatory label and 135 (3.4%) are NON_REGULATORY service complaints. The imbalance is a property of the source: the CFPB database predominantly receives complaints with a regulatory nexus, and the curation deliberately keeps the natural mix rather than rebalancing, so stage-1 metrics reflect production-like prevalence. This is why PR-AUC (not accuracy or ROC-AUC) is the primary stage-1 metric, why the classifiers use class weighting (`class_weight='balanced'; scale_pos_weight`), and why the score-distribution and calibration figures in the development document matter more than headline accuracy.

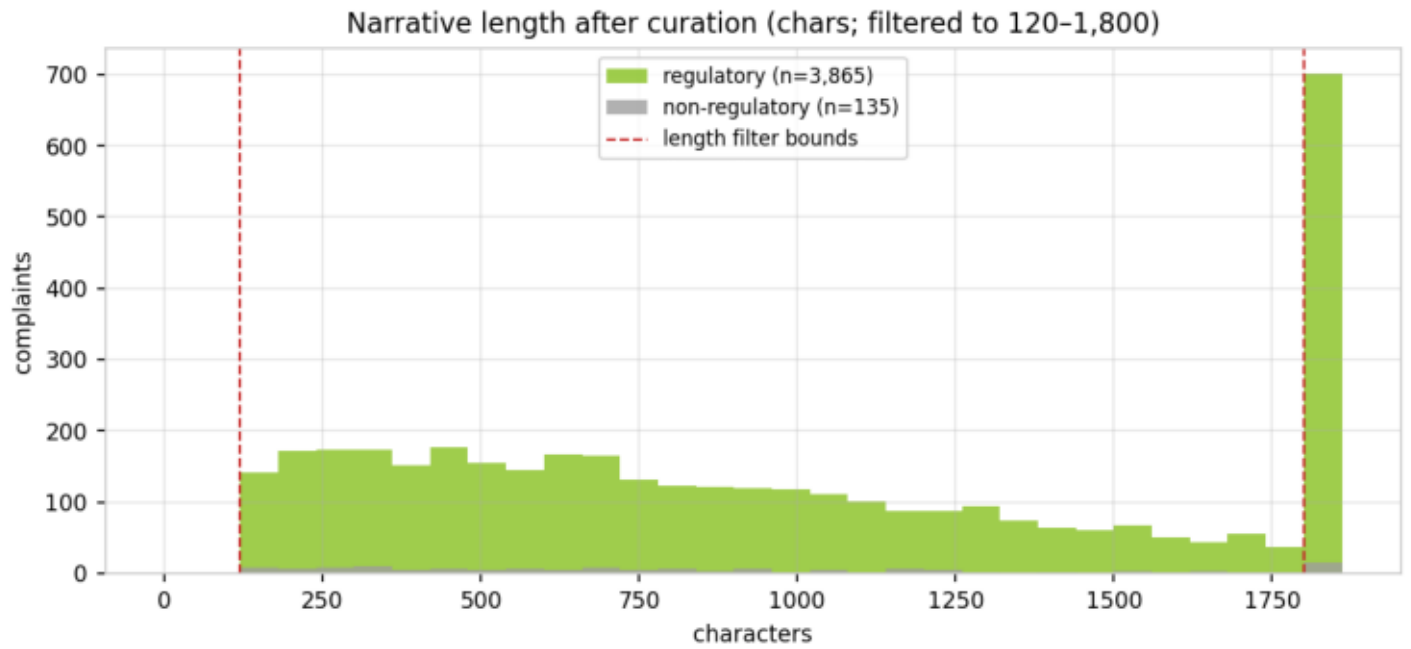
3 · Product mix



4 · Label coverage

label	category	n	share
UDAAP	UDAAP — Unfair, Deceptive	680	17.0%
LOAN_SERVICING	Loan Servicing (Auto/Student)	472	11.8%
REG_E_UNAUTHORIZED	Reg E / EFTA — Unauthorized	276	6.9%
BSA_AML	BSA/AML — Account Freeze	216	5.4%
TILA_ORIGINATION	TILA — Credit Origination &	213	5.3%
FDCPA_THREATS	FDCPA — False Statements	208	5.2%
FDCPA_DEBT_VALIDATION	FDCPA — Debt Validation /	207	5.2%
REG_E_ERROR_RESOLUTION	Reg E / EFTA — Error Resol	197	4.9%
REG_Z_DISCLOSURE	Reg Z / TILA — Fees, Inter	196	4.9%
FCRA_ACCURACY	FCRA — Accuracy of Repor	179	4.5%
RESPA_SERVICING	RESPA — Mortgage Servi	168	4.2%
REG_Z_BILLING	Reg Z / FCBA — Billing Erro	156	3.9%
FDCPA_COMMUNICATION	FDCPA — Communication	138	3.5%
GLBA_PRIVACY	GLBA / Privacy — Informat	136	3.4%
NON_REGULATORY	Non-Regulatory — General	135	3.4%
RESPA_LOSS_MITIGATION	RESPA — Loss Mitigation &	109	2.7%
FCRA_INVESTIGATION	FCRA — Reinvestigation of	107	2.7%
ECOA_DISCRIMINATION	ECOA / Reg B — Credit Dis	61	1.5%
FCRA_PERMISSIBLE_PURPOSE	FCRA — Permissible Purpos	48	1.2%
TISA_REG_DD	TISA / Reg DD — Deposit A	36	0.9%
REG_CC_FUNDS	Reg CC — Funds Availabilit	26	0.7%
SCRA_MLA	SCRA / MLA — Servicemen	13	0.3%
SALES_PRACTICES	Sales Practices — Unautho	12	0.3%
ECOA_ADVERSE_ACTION	ECOA / Reg B — Adverse A	11	0.3%

5 · Narrative length



6 · Train / validation / test design

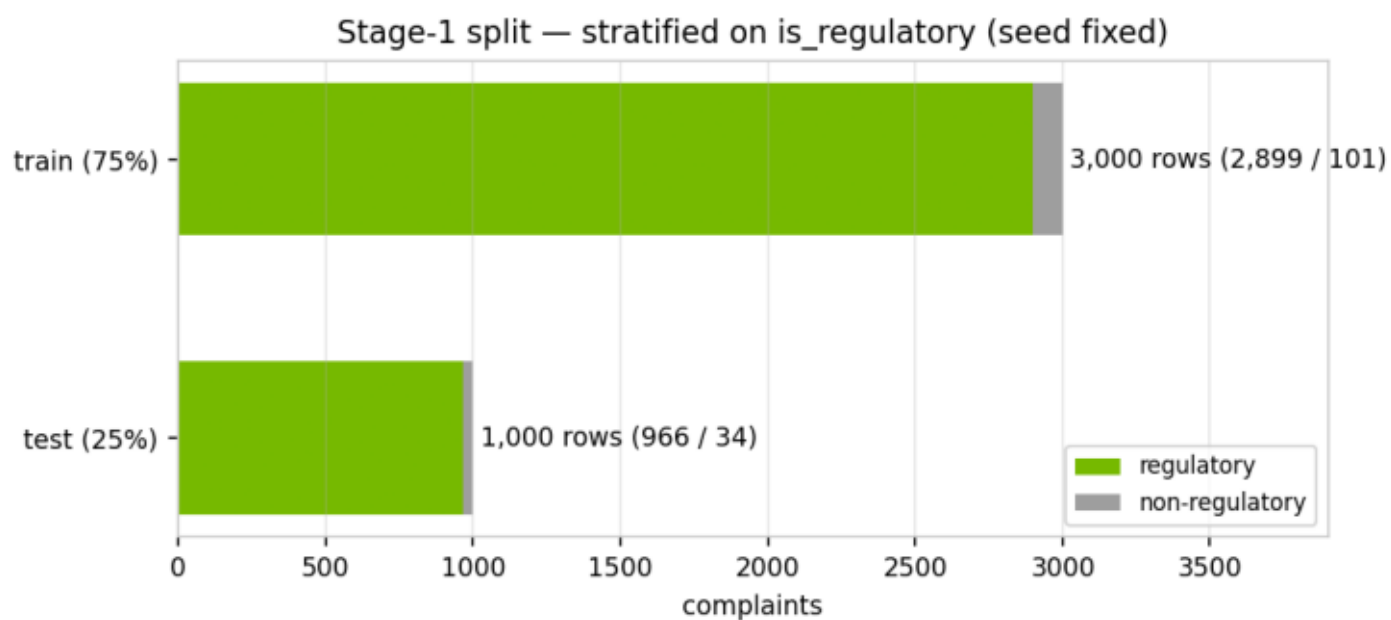
Stage 1 uses a stratified 75/25 train/test split (fixed seed). There is no separate validation split: with only two candidate models and a single primary metric (PR-AUC), the held-out test set doubles as the selection set. This is documented as a limitation – champion selection on the test set risks selection optimism as the candidate pool grows; the validation report's golden-set condition and any move to hyperparameter search would require a dedicated validation fold (60/20/20) or cross-validation.

Stage 2 involves no training: it is a prompted LLM over retrieved context. Its evaluation uses a separate stratified sample (n=115) of regulatory complaints scored against the weak labels, reported in the validation report. No complaint text is used to fit stage-2 parameters.

6 · Split composition

split	rows	regulatory	non-regulatory	reg rate
train	3,000	2,899	101	96.63%
test	1,000	966	34	96.60%

6 · Split (stratified)



7 · Data-quality checks (recomputed)

check	result	expectation
rows	4,000	= TARGET_ROWS (4,000)
missing values (key columns)	0	complaint_id, product, issue, narrative
exact duplicate narratives	0	post-normalization
length bounds respected	min 120 · max 1800	filter is 120–1,800 chars
PII masking present	82.3% of narratives	CFPB 'XXXX' masking tokens
max complaints per product-issue	126	cap is 350
label coverage	24 of 24 categories	all categories populated
regulatory / non-regulatory	3,865 / 135	96.6% regulatory